LUT University Autumn 2025
BK10A6400 Basics of FE Analysis
Marko Matikainen

Bonus task 6
Thick beam structure analyzed with two different two-dimensional four-node bilinear elements
Possibility to earn five bonus points
Deadline: Before the lectures on the 1st of December

Your task is to derive a two-dimensional quadrilateral four-node bilinear solid element (plane element "Q4") based on **a selective reduced integration scheme** via MATLAB or Python symbolic toolbox and solve displacements at the free end using a four-node bilinear solid element (plane element) based on full Gaussian integration and selective reduced Gaussian integration schemes with various finite element meshes. Furthermore, you need to analyse the rate of convergence (displacement (or error) vs DOFs) of a beam structure using four-node bilinear solid element (plane element) based on full Gaussian integration and selective reduced Gaussian integration schemes with various finite element meshes. Consider displacements in the y-direction at the beam's free end, compute the average, and draw convergence curves (displacement in the y-axis and DOFs in the x-axis) for both elements using meshes (4x1,8x2,16x4,32x8,64x16). Explain what can be seen from convergence curves and why elements produce different convergence curves.

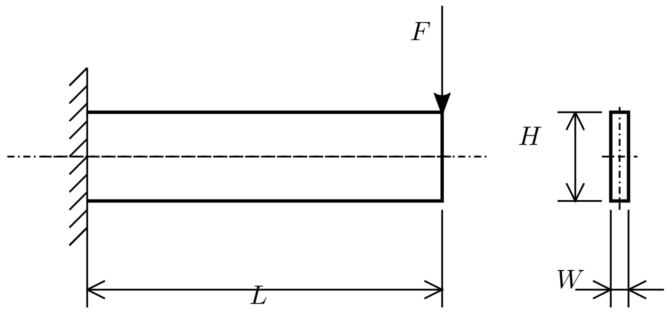

Figure 1. Thick cantilever problem

The parameters of the problem are given in Table.

Force, F [N]	Young's mod., E [Pa]	Poisson's ratio, ν	Length, L [m]	Height, H [m]	Width, W [m]
62500	$2.07 \cdot 10^{11}$	0.3	2	0.5	0.1

Write a small report that presents global displacements (average) at the free end explicitly in the table with various meshes, and present convergence curves. Submit the report in PDF format. Also, submit all necessary MATLAB/Python files (m-files/py-files) and verify that they run without issues. Bonus points will be shared based on the MATLAB/Python solution as follows: correct derivation of the element with reduced integration scheme (1 point), solved displacements (average) in tables with both elements (1 point), convergence curves with both elements (2 points), and correct explanation for the difference between elements (1 point). Please note that no points without a clear report (PDF format!) and MATLAB/Python codes (m-files!).

Code

All the necessary codes to perform static analysis of the cantilever problem are provided to solve the problem. You need to make a small code that presents convergence curves.

run_DeriveQ4xi_FullGauss.m.py	run firstly to export stiffness matrix
run_DeriveQ4psR_v12.m/.py	modify numerical integration and run to export stiffness matrix
run_LinStatic_Q4_v20.m/.py	Solves cantilever problem
RectangularMeshQ4.m/.py	Makes list of nodal positions and element's nodal connectivity
ElemDOFsConnectivity_Q4.m/.py	Makes DOF's connectivity from nodal connectivity
DofID.m/.py	Returns DOF ID with given nodes and nodal DOFs

Convergence curves

To produce convergence curves, you can estimate vertical displacement error between FE solution and analytical solution from beam theory as $Error = |v_{FEM} - v_{max}|$. Present error in the y-axis and DOF's in the x-axis.

TIPS

To get rid of shear locking (of bilinear plate element), you should make separate integration for energies related to shear and normal deformations.

$$U = U_n + U_s \tag{1}$$

$$U_n = \frac{1}{2} l_z \int_V \underbrace{\boldsymbol{\sigma}_n^T \boldsymbol{\varepsilon}_n \det(\boldsymbol{J}_e)}_{f_n(\xi,\eta)} \, \mathrm{d}\boldsymbol{\xi} = \frac{1}{2} l_z \int_V \boldsymbol{\varepsilon}_n^T \boldsymbol{D}_n \boldsymbol{\varepsilon}_n \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} \tag{2}$$

$$U_s = \frac{1}{2} l_z \int_V \underbrace{\sigma_{xy} \gamma_{xy} \det(\boldsymbol{J}_e)}_{f_s(\xi,\eta)} \, \mathrm{d}\boldsymbol{\xi} = \frac{1}{2} l_z \int_V D_s \gamma_{xy}^2 \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} \tag{3}$$

Note that $f_s(\xi, \eta)$ and $f_s(\xi, \eta)$ are integrands, and the number of integration points is estimated based on those functions.

Analytic solution based on beam theory

$$\frac{d^2 v}{dx^2} = -\frac{M(x)}{EI(x)} - \frac{\kappa_{xy} q(x)}{GA} \tag{4}$$

Maximum deflection for the cantilever beam under applied force at the free end can be found as:

$$v_{max} = \underbrace{\frac{FL^3}{3EI}}_{\text{bending}} + \kappa_{xy} \underbrace{\frac{FL}{GA}}_{\text{shear}} \tag{5}$$

where κ_{xy} is a shear correction factor (needed to use in the beams based on the first-order shear deformation theory (FSDT)). It can be approximated to be $6/5$ for a rectangular cross-section.